Sai Ganesh Chatharasupalli

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M.Tech Computer Science
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Github | **LinkedIn**

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech (CSE)	Indian Institute of Technology, Guwahati	8.11(Current)	2025-Present
B.Tech (CSE)	KKR AND KSR Institute Of Technology And Sciences	8.64	2021-2025
Intermediate	The Board of Intermediate Education, Andhra Pradesh	92.4%	2019-2021
10th Class	Board of Secondary Education, Andhra Pradesh	87.4%	2019

Internship

- **Amazon** *May 2026 – Jul. 2026*
  Software Development Engineer (SDE) Intern Hyderabad, India
  – Engineered a Lambda event pipeline using S3, SQS, and DLQ to process 250+ resolution evaluation signals and publish events downstream **within minutes** instead of daily ETL jobs.
  – Built event-driven, multi-stage, multi-region **AWS infrastructure** in CDK using **VPC**, PrivateLink, monitoring, and canary deployments across 4 regions, with Transitive Auth for downstream Coral service calls.
  – Shipped 15 CRs across 14 packages and 3 interconnected services (GR, CCS, CRS), spanning **2,686 lines** across service integration, GraphQL schemas, AWS infrastructure, and event-processing components.

Projects

- **CDP Event Publishing Pipeline** *May 2026 – Jul. 2026*
  Amazon | AWS, Java, TypeScript, GraphQL
  – Delivered the **Resolution Evaluation Event Processor** by defining CDP event schemas in GraphQL for customer, purchase, returns, and concession-history signals, along with resolution evaluation details.
  – Integrated the AWS event-processing flow using SQS, Lambda, S3, and DLQ with the **ABCDA pattern**, incorporating Handler and Manager layers, Dagger, JSON parsing, and per-purchase-unit processing.
- **WalletPay – Digital Wallet Platform** *2025*
  Spring Boot, React, PostgreSQL, Docker, JWT **GitHub**
  – Built a full-stack digital wallet with JWT authentication, supporting wallet management, fund deposits, P2P transfers, and persistent transaction tracking with unique transfer identifiers and real-time transaction updates.
  – Architected **Controller-Service-Repository** architecture with Spring Boot, JPA, Hibernate, and PostgreSQL, with Dockerized database and validated transaction workflows.
- **Distributed Game Server – Chunk Based Map Partitioning** *Oct. 2025*
  Go, UDP, HTTP, React, Three.js **GitHub**
  – Engineered a distributed multiplayer game server that partitions the world into spatial chunks, coordinates chunk ownership across game-server instances, and handles player movement and cross-server migration.
  – Implemented **UDP-based server communication**, **chunk state synchronization**, dynamic chunk loading, and real-time world updates through a React/Three.js client.
- **PeerVault – P2P Distributed File System** *Aug. 2026*
  Python, FUSE, TCP/IP, UDP, Multithreading, Socket Programming **GitHub**
  – Built a P2P distributed storage system with **UDP peer discovery** and FUSE-based mounting of remote storage.
  – Implemented **TCP file operations**, metadata caching, multithreading, XOR encryption, and path validation.
- **Task-Aware Adaptive Image Compression** *Sept. 2026 – Ongoing*
  Python, PyTorch, ResNet, Machine Learning | MTP under Jatindra Kumar Deka **GitHub**
  – Developing a task-aware image compression framework that allocates image-specific compression levels based on learned image importance under a fixed storage budget.
  – Achieved **80.07% validation accuracy** with ResNet-18 under image-specific compression levels.

Technical Skills

- **Programming Languages**: Java, C++, C, Python
- **Web Technologies & Databases**: Spring Boot, React, HTML, CSS, SQL, MongoDB
- **Cloud & Infrastructure**: AWS, Lambda, S3, SQS, DLQ, VPC, PrivateLink

Key courses taken

- **Core & Electives**: Data Structures and Algorithms, Concurrency Theory, Distributed Systems, Speech Processing
- **Online Courses**: Master Class on Java Programming (Udemy)

Achievements

- **Competitive Programming**, LeetCode | CodeChef
  – LeetCode Knight Badge | Max Rating: 1870 | Rank 599 in Leetcode Weekly Contest 516 | ID: **saiganesh0543**
  – CodeChef 3⋆ Coder | ID: **saiganesh_18**
- **NPTEL Elite Topper**, Elite Topper (Top 1%) in *Algorithmic Graph Theory and Data Structures* *2026*